

CORNERSTONE Off-the-Shelf Demonstration Chip Guidelines

I. Chip overview

The SiN300 off-the-shelf demonstration chips will fabricate on the standard CORNERSTONE Photonics Innovation center (C-PIC) silicon nitride platform with a 2 μm SiO₂ top cladding and a 3 μm buried oxide (BOX) layer. Metal heaters were integrated as thermo-optic phase shifters. Most of the devices work at 1550 nm wavelength and if it is not specified, the working wavelength is 1550 nm when these devices are discussed.

The layout of this demonstration cell is shown in Fig. 1.

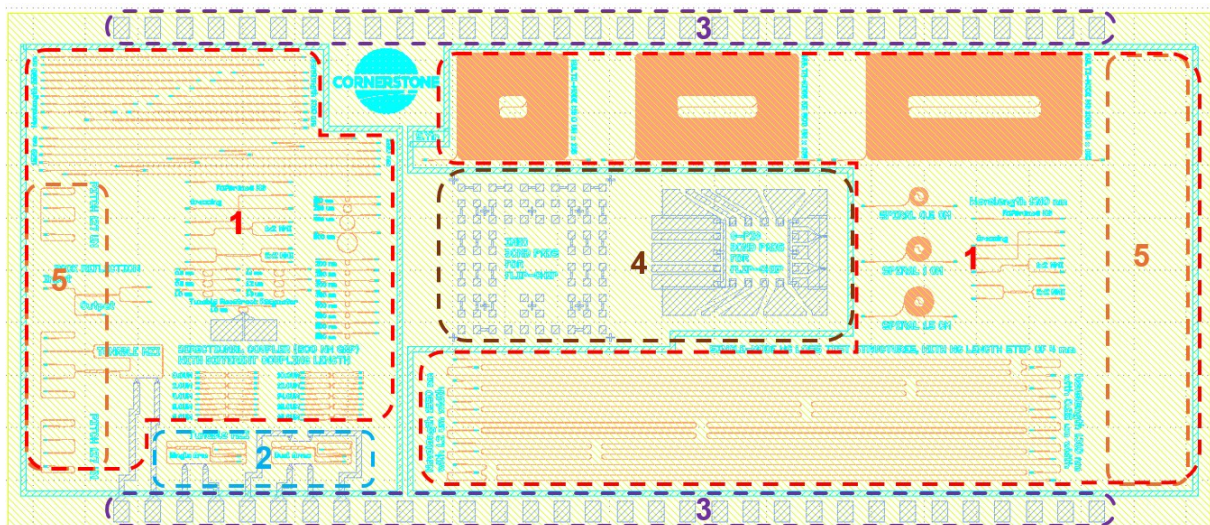


Figure 1. Layout overview

Generally, there are five groups of devices/structures and they are:

1. Photonic devices based on strip waveguide
2. Active component: Mach-Zehnder interferometers
3. Electrical contact pads for wire bonding
4. Electrical contact pads for flip chip bonding
5. Grating coupler arrays for optical fiber bonding

Each component is labelled physically on the chip, which is visible under a microscope.

II. Photonic devices based on strip waveguides

1. Basic components

The most basic components in a photonic integrated circuit (PIC) are straight waveguides, waveguides bends, waveguide crossings, and beam splitters/couplers.

As shown in Fig. 2, there are four devices circled, (1) is a short straight waveguide in between two sets of grating coupler and taper for light input and output, (2) a waveguide crossing with four input/output grating couplers, (3) a 1×2 multi-mode interferometer (MMI) beam splitter/coupler with input/output grating couplers, and (4) a 2×2 MMI beam splitter/coupler with input/output grating couplers.

In device (1), the optical loss of the short straight waveguide is negligible and the transmission of the device reflects the transmission of the two grating couplers. A peak transmission of -20 to -24.5 dB is expected.

In device (2), for example, when the light is launched into the device from **port 1**, a peak transmission of -20 to -24.5 dB is expected from **port 4** because the loss the crossing is insignificant, whilst the transmission from **port 2** and **port 3** should be negligible.

In device (3), assuming the light is launched into the device from the left port, similar spectra are expected from the two ports on the rightm which means the 1×2 MMI can evenly split the optical power.

In device (4), which is similar to (3), assuming the light is launched into the device from one of the two left ports, similar spectra are expected from the two ports on the right, which means the 2×2 MMI can also evenly split the optical power. Please note: it is difficult to realize a exactly even splitting using a 2×2 MMI especially in a wide wavelength range. Also, there is $\pi/2$ phase difference on the output ports.

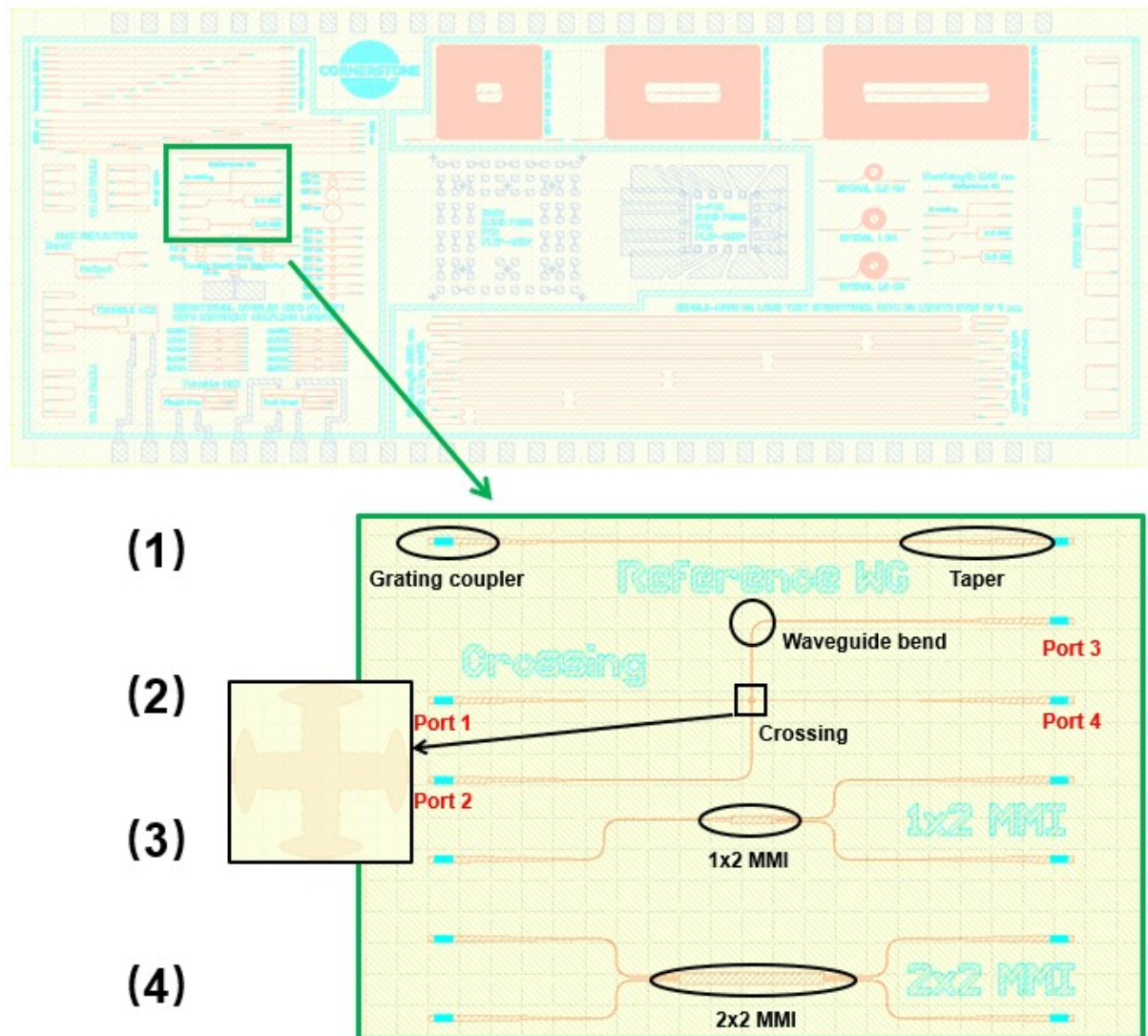


Figure 2. Basic components

Please note that the straight waveguide in this section and following sections (if not specified) have a width of $1.2 \mu\text{m}$ and they are single mode (SM) transverse-electric (TE) waveguides. The waveguide bends also have a width of $1.2 \mu\text{m}$. All the bends used on this demonstration chip are arc bends and thus it has to support single mode only.

2. Tunable MZI

Using two MMIs one can easily build a Mach-Zehnder interferometer (MZI). See the device in Fig. 3 as an example of MZIs, which demonstrates a single-arm controlled MZI (left device) and a dual-arms controlled

MZI (right device). Each of them is formed by a 1×2 MMI, 2×2 MMI, straight waveguides, waveguide bends and grating couplers. On the transmission spectrum, one should observe periodic constructive and destructive interference. The period is the so-called free-spectral-range (FSR), which is correlated with the optical path difference of the two MZI arms between the 1×2 MMI and 2×2 MMI.

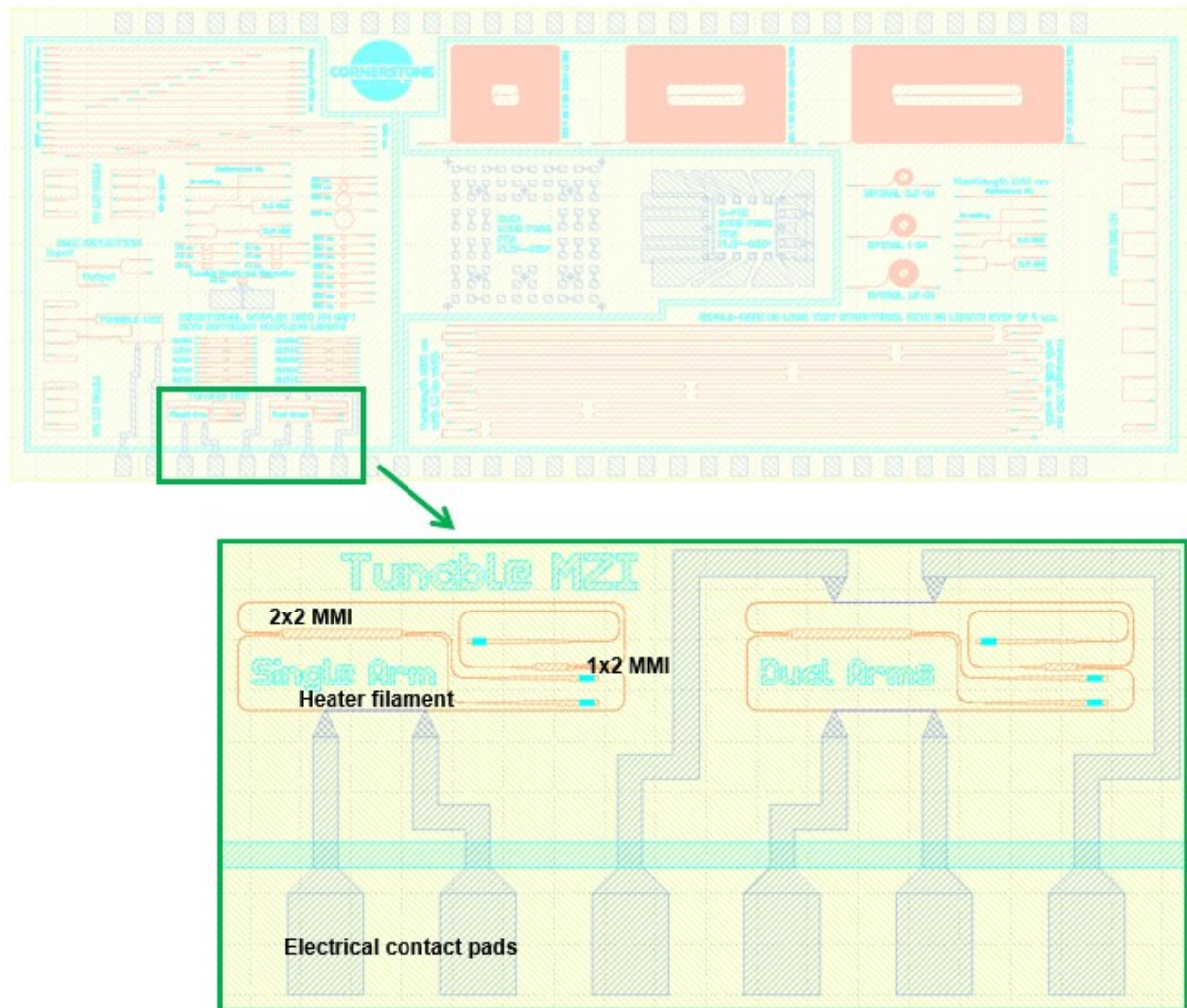


Figure 3. Tunable MZI

Please note that a thermo-optic phase shifter is integrated on one of the MZI arms (left device). When a voltage difference is applied to the two contact pads, there will be a current flow through the filament to heat the MZI arm up, eventually resulting in the effective arm length different change. Thus when different voltages are applied, the constructive/destructive wavelength and the FSR can be tuned. Or, if the input wavelength is fixed and the output optical power is monitored, when the heating power is linearly increased, the optical power should vary as a sinusoidal curve. The two heater filaments design, on the right device, provides an additional control to balance the thermal crosstalk as a dummy arm. Please note that contact pads are designed for probes, whilst they can be bonded as well.

3. Grating coupler back-reflection

Back-reflection is also a key parameter of a grating coupler design, which may induce unexpected system noise for an on-chip built detection system by using a grating coupler as an output terminal. As shown in Fig. 4, a group of 4 grating couplers with a 2×2 MMI is used to detect the back-reflection of the grating coupler. Input and output terminals are elaborated in the on-chip marks, while the remaining two grating couplers serve as references with different path lengths ($\Delta L = 10 \mu\text{m}$) to the MMI. The whole structure functions as a Michelson interferometer, which can monitor the back-reflected signal. If one

measure from the output terminal, one can obtain a sinusoidal curve on an oscilloscope.

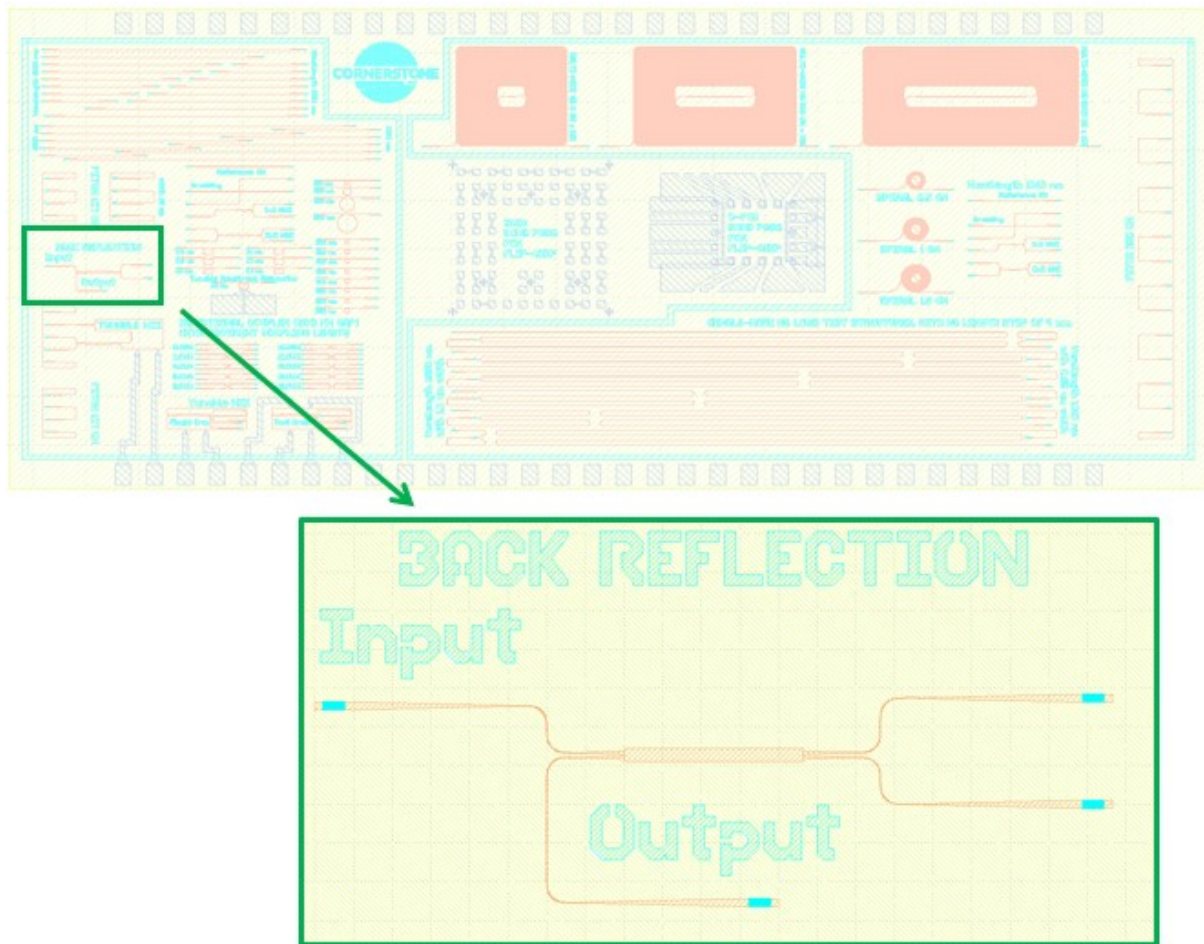


Figure 4. Grating coupler back-reflection test

4. Directional couplers

As shown in Fig. 5, there are a series of directional couplers with various coupling lengths. When two waveguides are placed in parallel and close enough each other, they are coupled evanescently and the optical power will exchange between them. With a fixed coupling gap of 350 nm , the power exchange rate (power per unit distance) is fixed. By varying the coupling length, different amounts of optical power will be transferred from one waveguide to the other. So if the light is launched into the device from one of the grating couplers, the transmission ratio at the grating couplers on the other side will be different with different coupling length.

The directional coupler is also widely used in PICs. For example, it is the key part of a racetrack resonator.

5. Racetrack resonators

Ring/racetrack resonators are the most common resonant components used in PICs. Some examples of racetrack resonators are shown in Fig. 6. The top six racetrack resonators are fix components, with different coupler gaps. Periodic resonance dips, which contribute constructive interference in the resonator, are expected in the transmission spectrum. The period, that is FSR, is mainly determined by the circumference of the racetrack, whilst the coupling gap has slight effect on the FSR. On the other

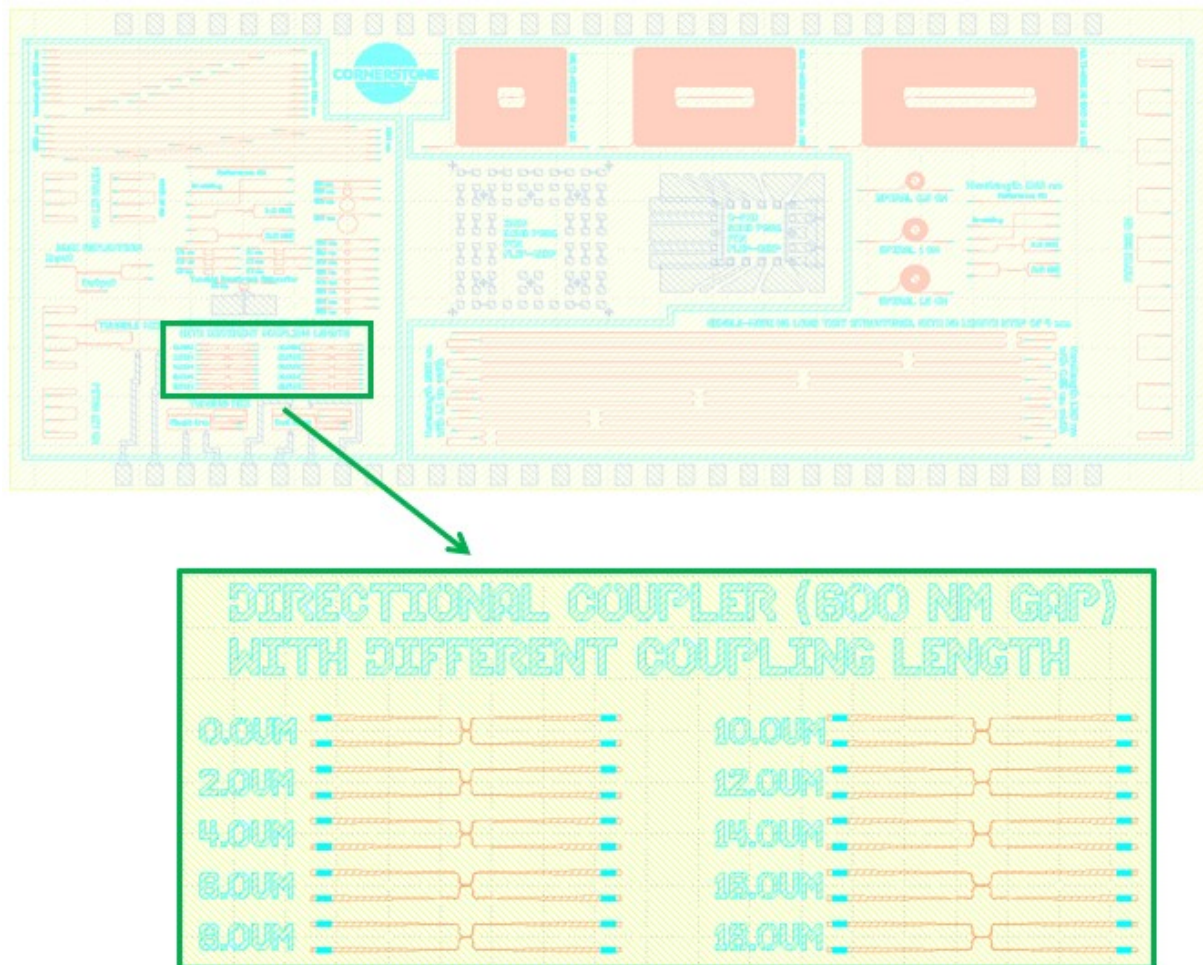


Figure 5. Directional couplers

hand, the coupling gap is a key factory in the design and it controls the Q factor and the coupling depth (extinction ratio) of the resonances.

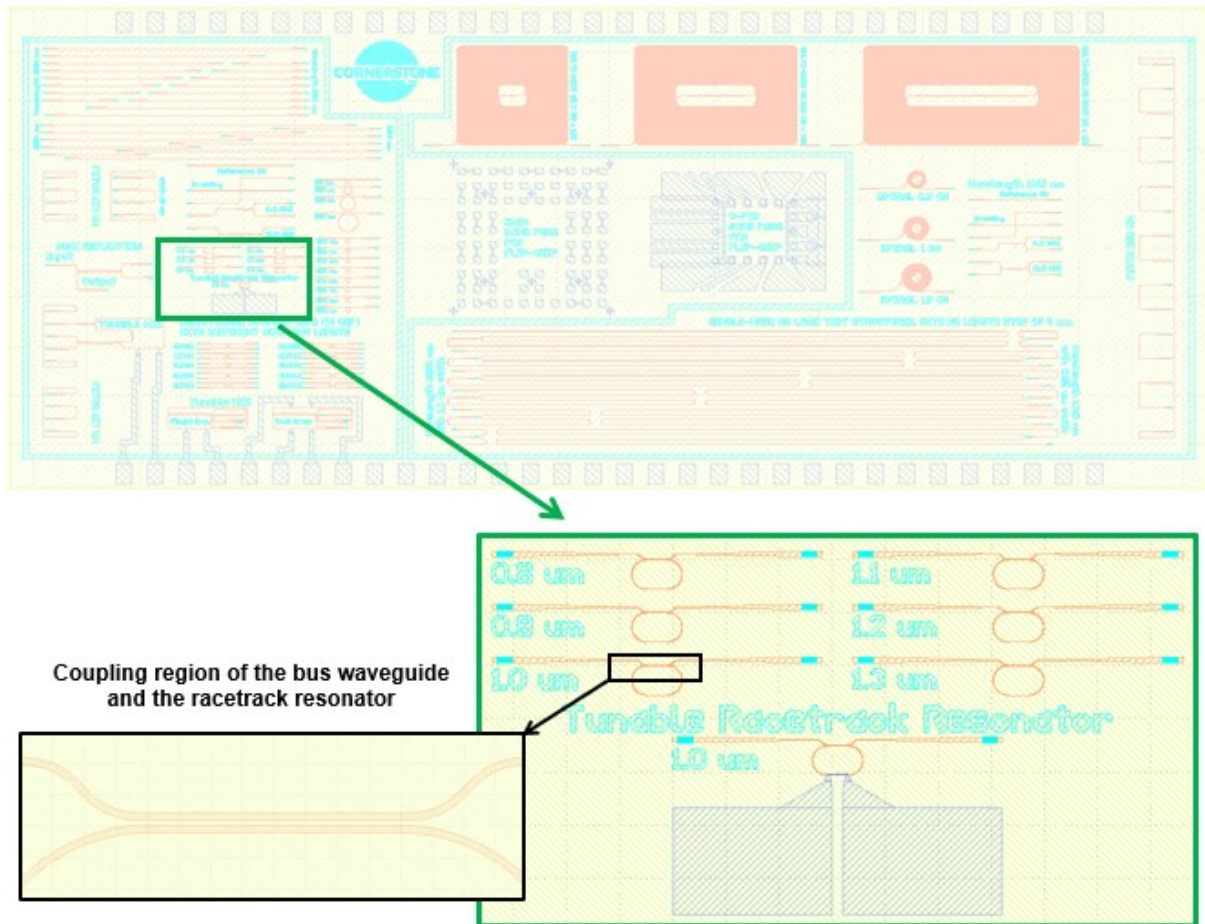


Figure 6. Racetrack resonators

The bottom racetrack resonator is a tunable design, in which a heater is built on the racetrack. When an electrical power is applied, the effective circumference of the racetrack will be increased and one should observe that the resonance will shift to the longer wavelength.

6. Ring resonators

Racetrack resonators are widely used because there is a long coupling region between the bus waveguide and the racetrack to guarantee the coupling coefficient. However, if a very large FSR and thus very small circumference are required, a racetrack design is not ideal. To minimize the circumference, one can consider ring resonator as shown in Fig. 7. To maintain a long enough coupling length between the bus waveguide and ring, bend couplers can be used as shown in Fig. 7. An annotation beside the ring resonator with a bend coupler records the perimeter of ring.

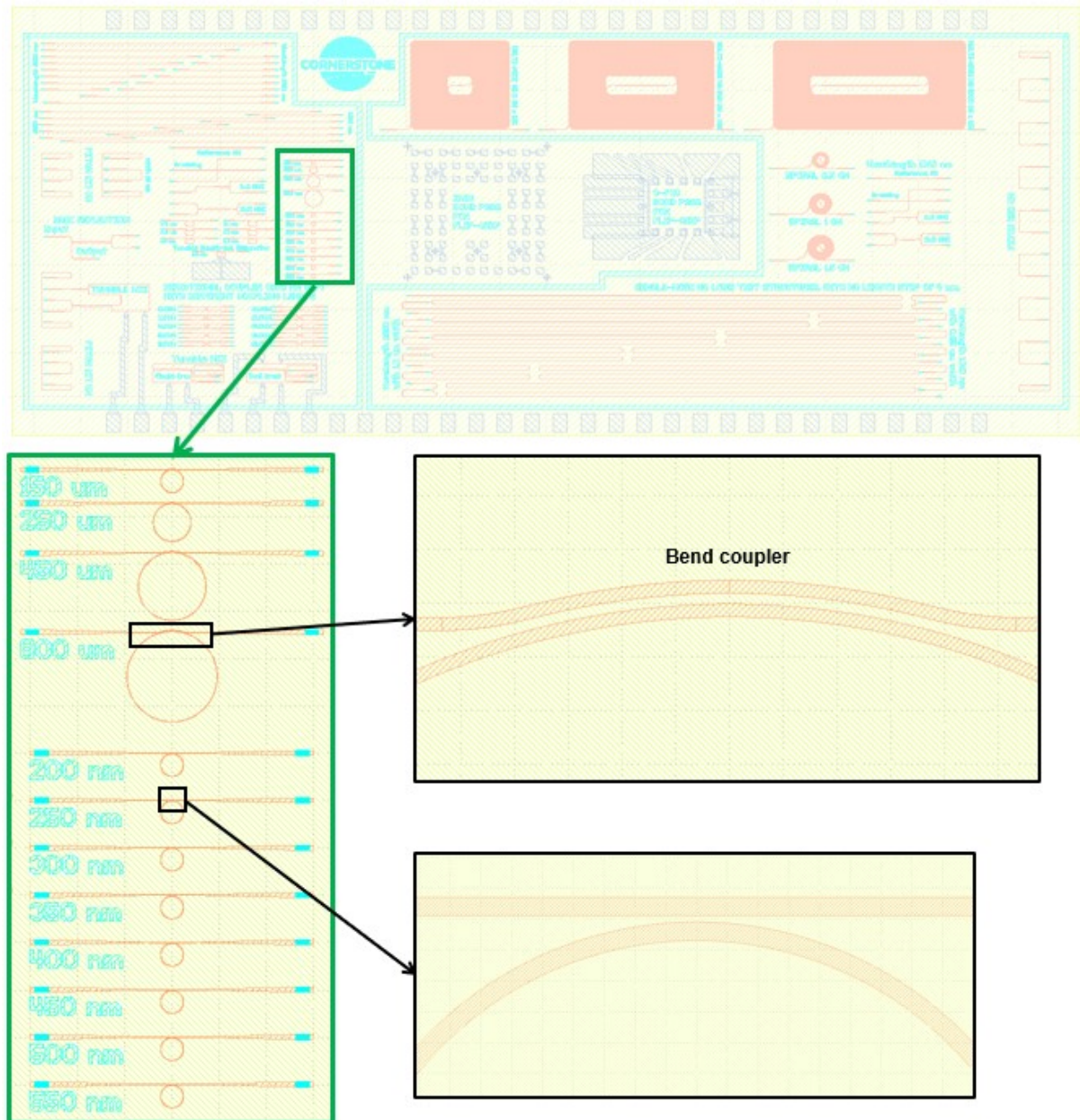


Figure 7. Ring resonators

A ring resonator with a straight bus waveguide is also demonstrated in Fig. 7, as shown in the bottom panel, with varying coupling gaps. The gap information shows near the corresponding design.

7. Spiral waveguides

As shown in Fig. 8, there are two types of spiral waveguide.

In design A, the straight part of the waveguides are multi-mode (MM) waveguides, with a width of $2 \mu\text{m}$, whilst the waveguide bend has to be SM to avoid mode crosstalk in the bend. They are connected using waveguide tapers as shown in Fig. 8. There are three sets of spiral waveguides in design A with stepped total waveguide length (optical path between the two grating couplers) and they can be used to measure the optical loss of the MM waveguide using the cutback method discussed above. Please note that the total waveguide length step is $500 \times 135 \mu\text{m}$ as labelled on the chip.

In design B, all the waveguides are arcs and thus SM waveguides. This is a popular design in sensing

applications.

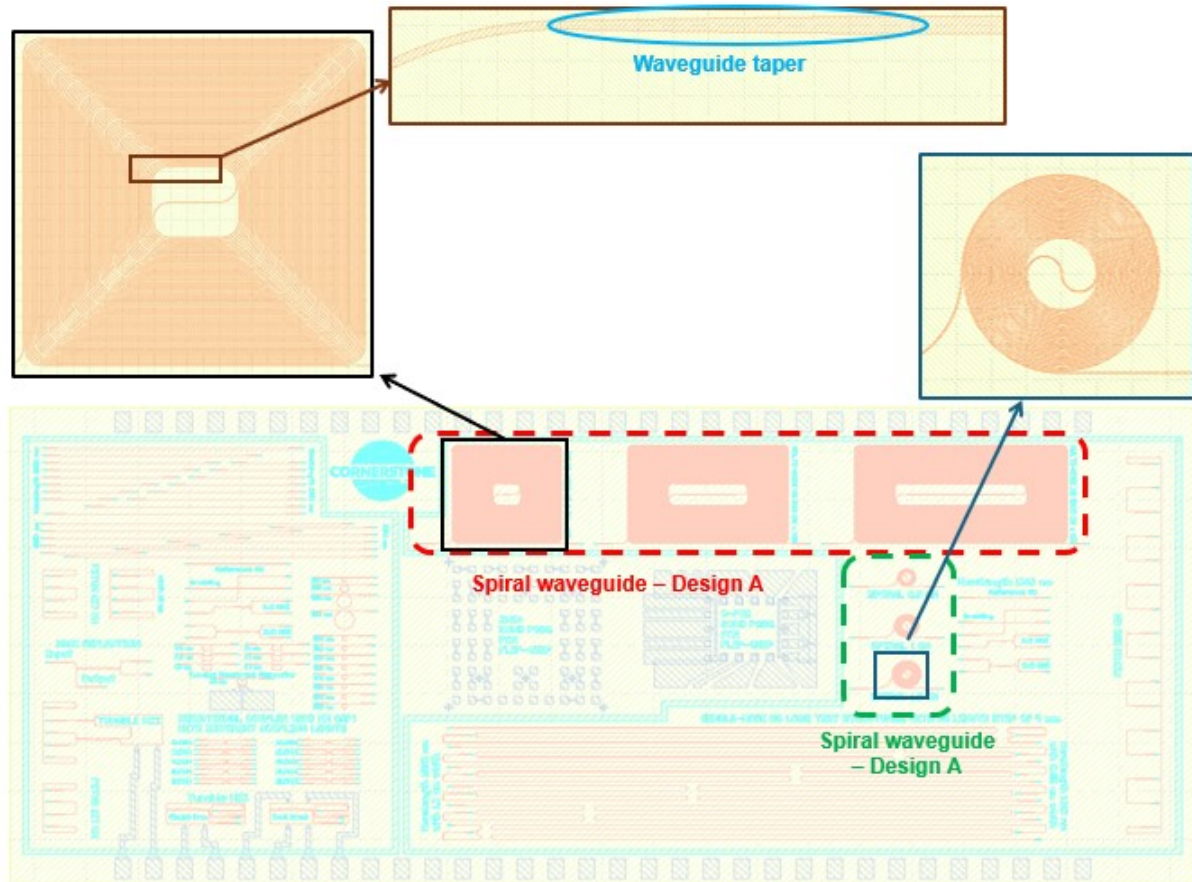


Figure 8. Spiral waveguides

III. Photonics devices for functional test

1. Crossing loss test structure

As demonstrated in Fig. 9, there are two sets of crossing propagation loss test structures for wavelengths of 1550 nm and 1310 nm , respectively. Each set contains 8 interconnected crossings with a different number of crossings. As shown in Fig. 9, two adjacent crossings are connected by a SM waveguide, and the structure has a periodicity of $20\text{ }\mu\text{m}$. The working wavelength is also elaborated in the on-chip marks. If one measures the transmission in any set, and then plots and linearly fits the transmission in dB against the relative crossing number, one can get the waveguide optical loss in dB per crossing unit due to the negligible waveguide loss. The distribution of the crossings set, i.e. the number of crossings in each set, is shown in the Table. 1.

Table 1. Crossings distribution

	Number of crossings							
	- From top to bottom ->							
1550 nm	100	80	60	50	40	30	20	10
1310 nm	10	20	30	40	50	60	80	100

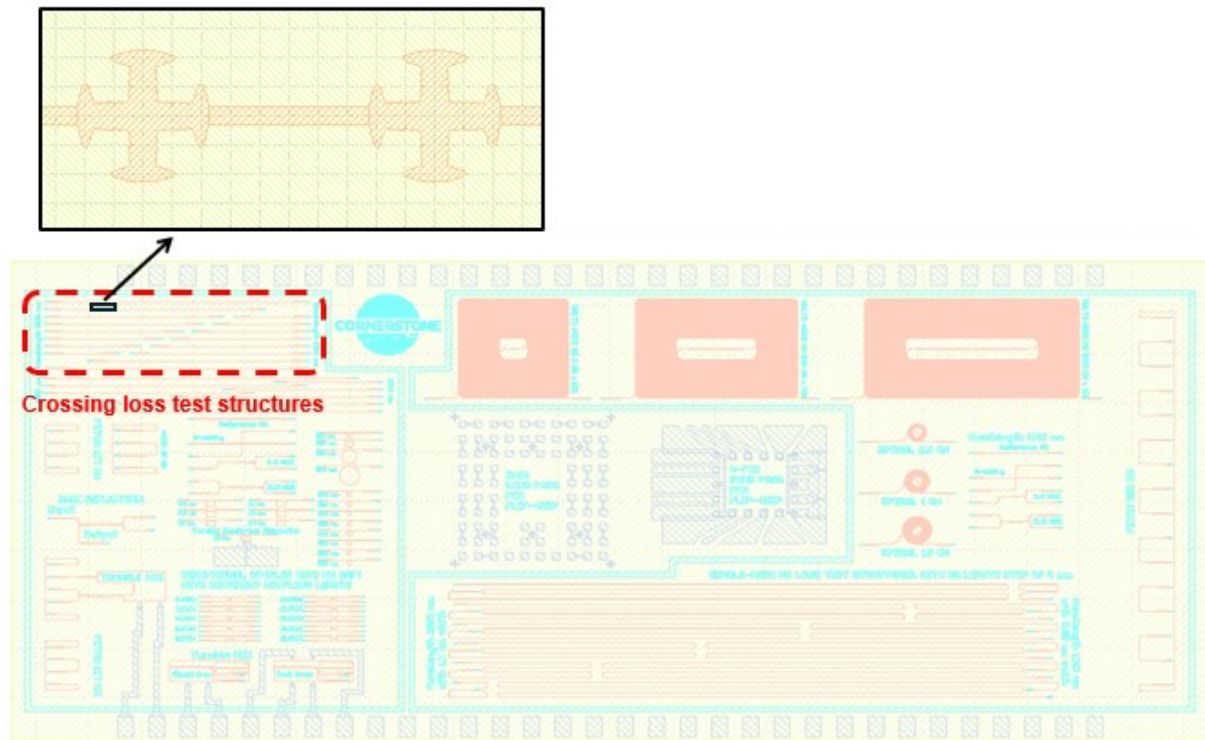


Figure 9. Crossing loss test structures

2. Short-taper loss test structure

In Fig. 10, Two sets of taper loss test structures for wavelengths of 1550 nm and 1310 nm , respectively. The taper is used to connect a SM waveguide and a MM waveguide, and its hyperbolic profile can reduce the coupling length between the two waveguides inside an optical circuit.

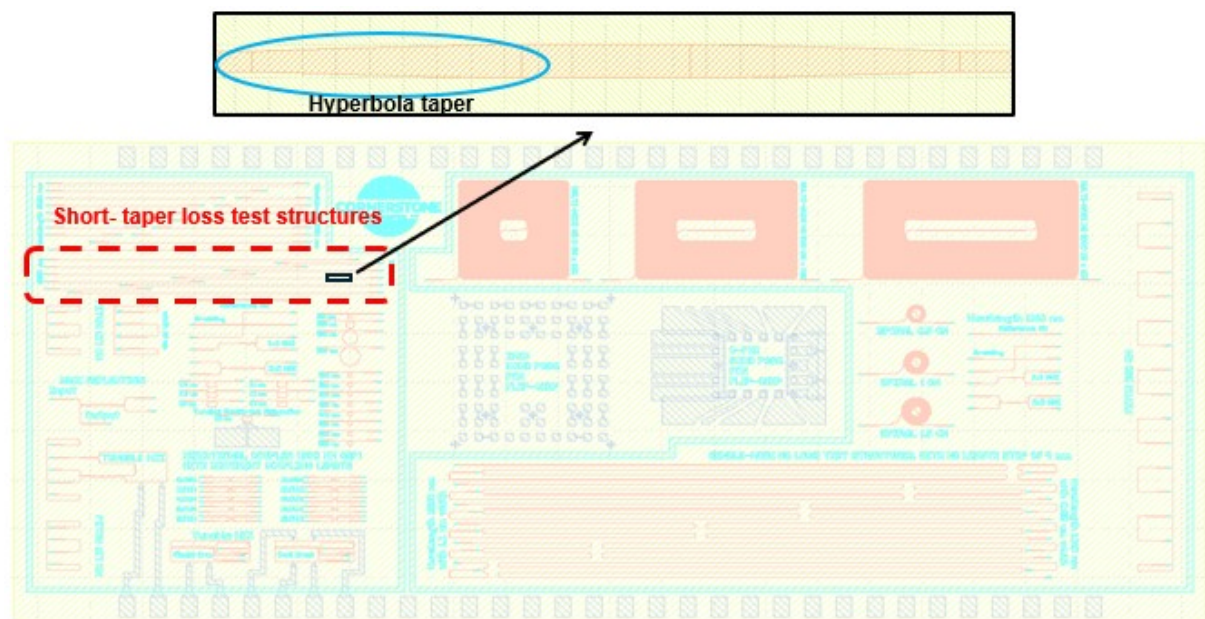


Figure 10. Short-taper loss test structures

In each set, there are 5 groups of hyperbola tapers, with the number of tapers increasing/decreasing in steps of 20. As shown in the top panel of Fig. 10, a pair of hyperbola tapers is used to convert a SM

waveguide to a MM waveguide and then back to SM. The wavelength is elaborated in the on-chip marks. By measuring the transmission in any given set and plotting it in dB against the relative taper number, a linear fit can be applied to determine the optical loss per unit in dB. The method is known as the cutback measurement.

3. Waveguide loss test structure

As marked in Fig. 11, there are two sets of waveguide optical loss test structures for wavelengths of 1550 nm and 1310 nm, respectively. In each set, there are 6 waveguides with a waveguide length step of 400 μm , which is given in the physical mark on the chip. The wavelength and waveguide dimensions are also elaborated in the on-chip marks. If one measures the transmission in any set, and then plots and linearly fits the transmission in dB against the relative wavelength length, one can get waveguide optical loss in dB per unit length. This is the so-called cutback measurement.

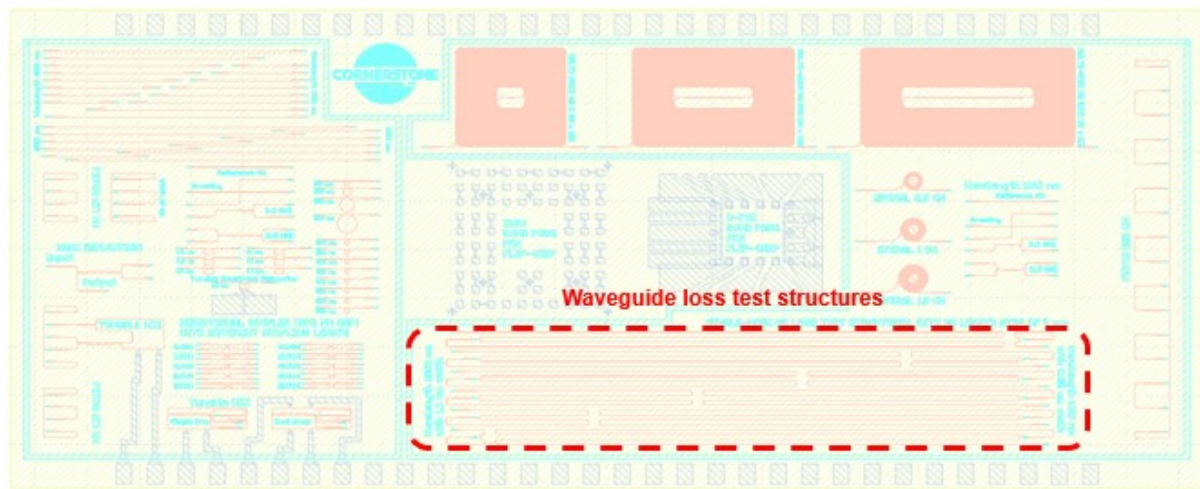


Figure 11. Waveguide loss test structures

IV. Optical packing - optical fiber bonding

For high-level integration, a PIC chip can be optically packaged. This PIC offers an example of optical packing: optical fibre bonding. As marked in Fig. 12, there are grating coupler arrays on both side of the PIC.

On the left side, the top and bottom arrays have four gratings and the middle one has six gratings. They all have a grating pitch of 1320 nm. Note that in the middle device, a tunable MZI is connected to the gratings and the heater filament is connected to the contact pads on the bottom edge of the PIC, which can be wire bonded to a PCB. This device is used to demonstrate full PIC integrating, including both optical packaging and electrical packaging.

On the right side, there is only one array, which has 16 gratings with a pitch of 250 μm .

In all of these grating coupler arrays, the first grating, the first grating and the last one are always connected directly using a so-called "optical shunt" to align the fibre array to the grating array during the fibre attach process, whilst the other gratings are functional gratings.

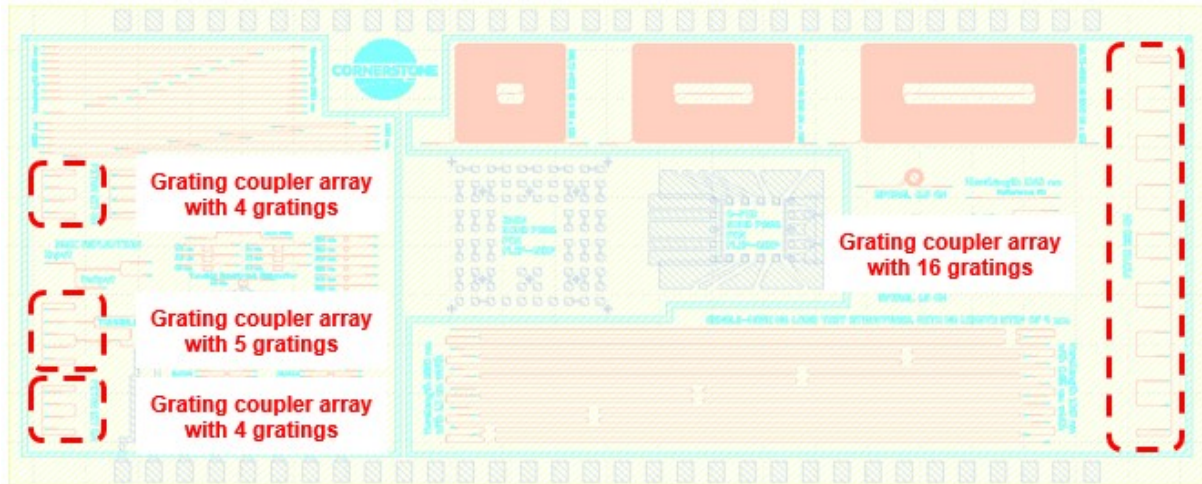


Figure 12. Grating coupler array template for optical fibre attachments

V. Electrical packaging

Two types of electrical packaging will be discussed: wire bonding and flip-chip bonding. Compared with wire bonding, flip-chip bonding can realize inter-chip electrical connections with ultra-small resistance because the contact pad are connected using sub-100 μm gold balls. Consequently, flip-chip bonding is a great choice to connect a high-speed optical modulator and its electrical driver.

1. Wire bonding

As marked in Fig. 13, there are two rows of metal contact pads on the top edge and bottom edge of the PIC, which are the templates for the CORNERSTONE wire bonding service, which matches the pad layout on the standard CORNERSTONE PCBs.

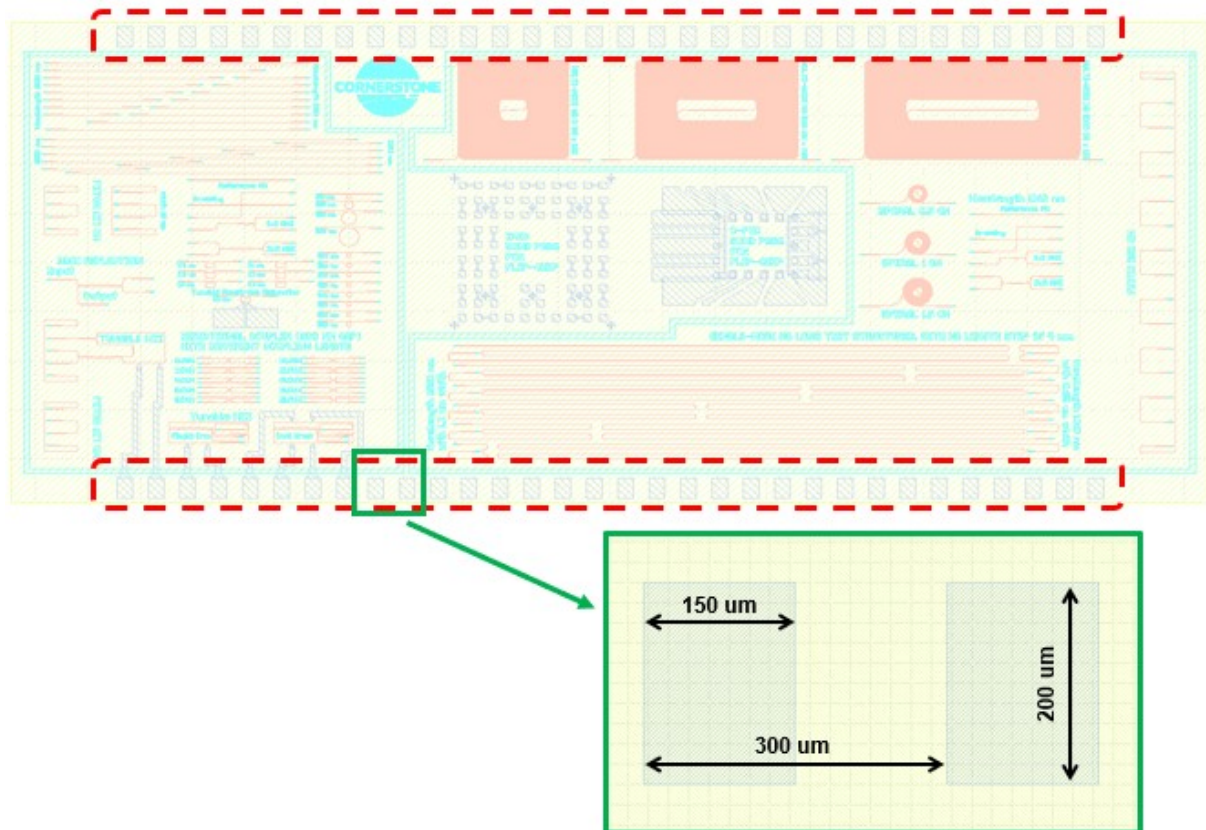


Figure 13. Metal contact pads template for wire bonding

2. Flip-chip bonding

There are two sets of flip-chip bonding pad templates, one to match the standard EUROPRACTICE IMEC design and the other one for the standard CORNERSTONE design, as shown in Fig. 14. One can measure the dimensions and the pitch of the pads in the GDS file.

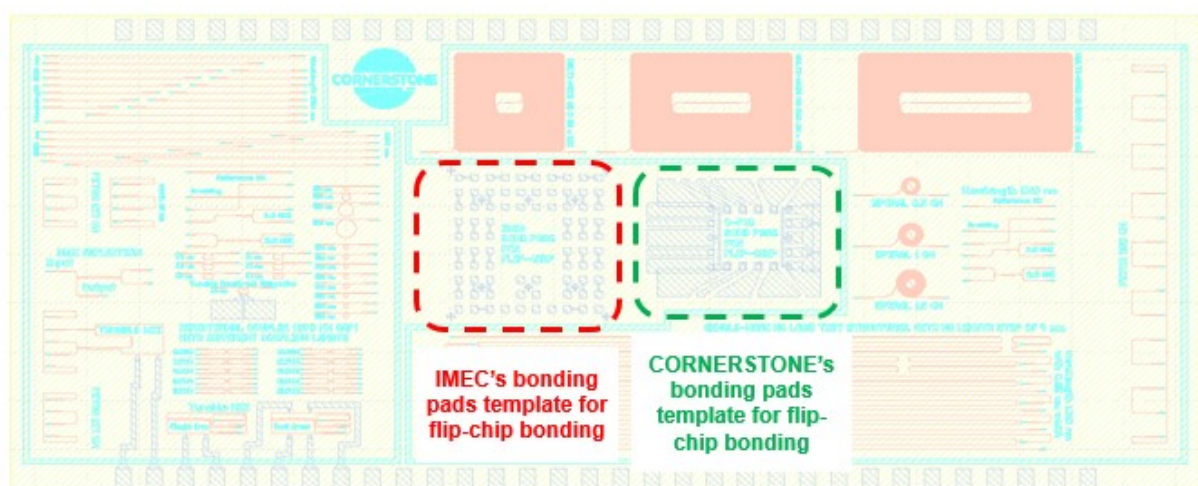


Figure 14. Contact pads template for flip-chip bonding